BIKES SALES

Queries & Insights

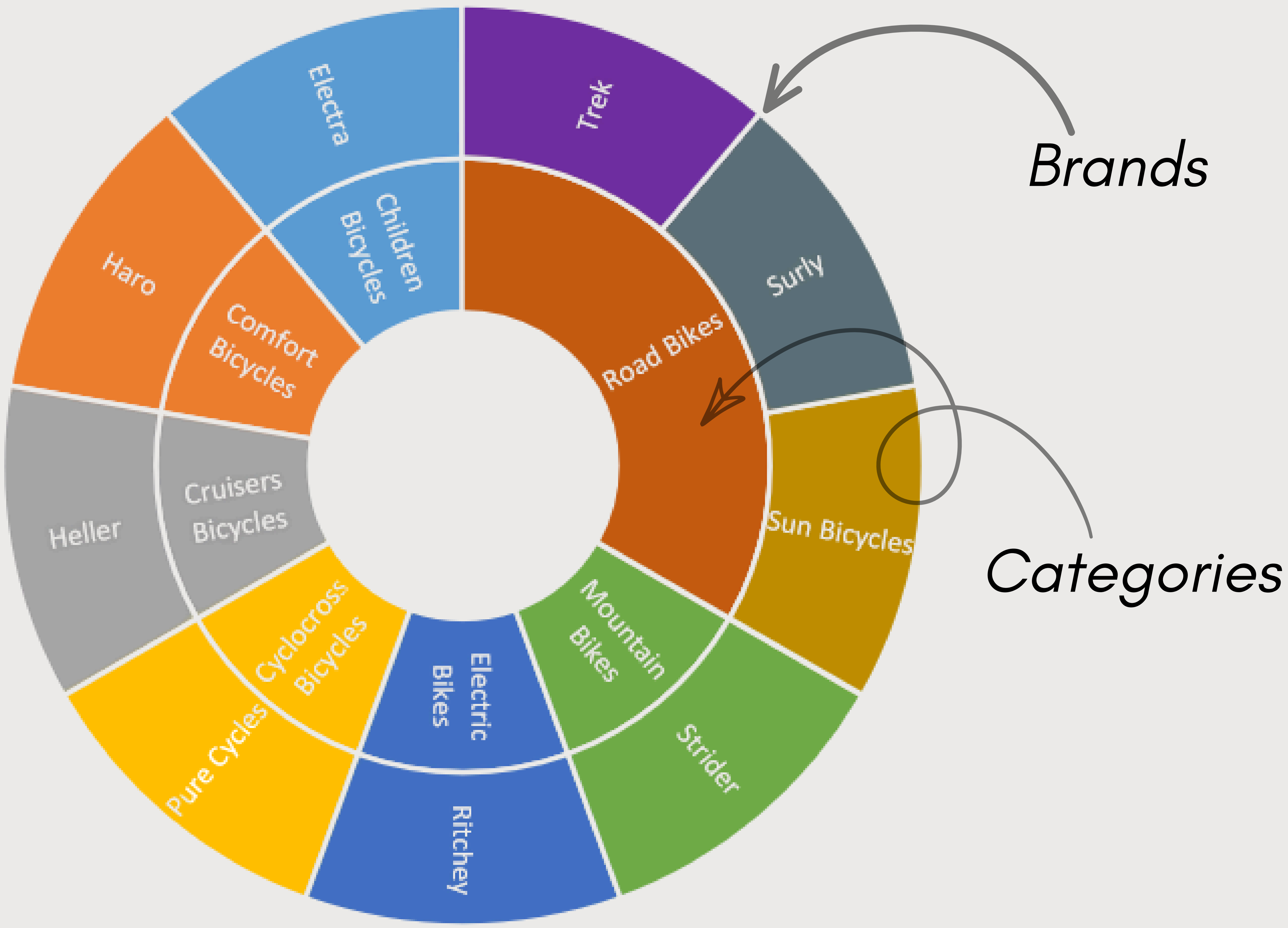
Using MySQL

Khushal Dusane

WHY ?

The objective of this analysis is to understand the business performance of the bike sales and support better decision-making using data. By analyzing customers, orders, products, brands, categories, and revenue trends, we can identify top-performing products, valuable customers, seasonal sales patterns, and areas where sales have declined. These insights help management improve marketing strategies, optimize inventory, increase revenue, and make informed business decisions based on data rather than assumptions.

WHAT ?



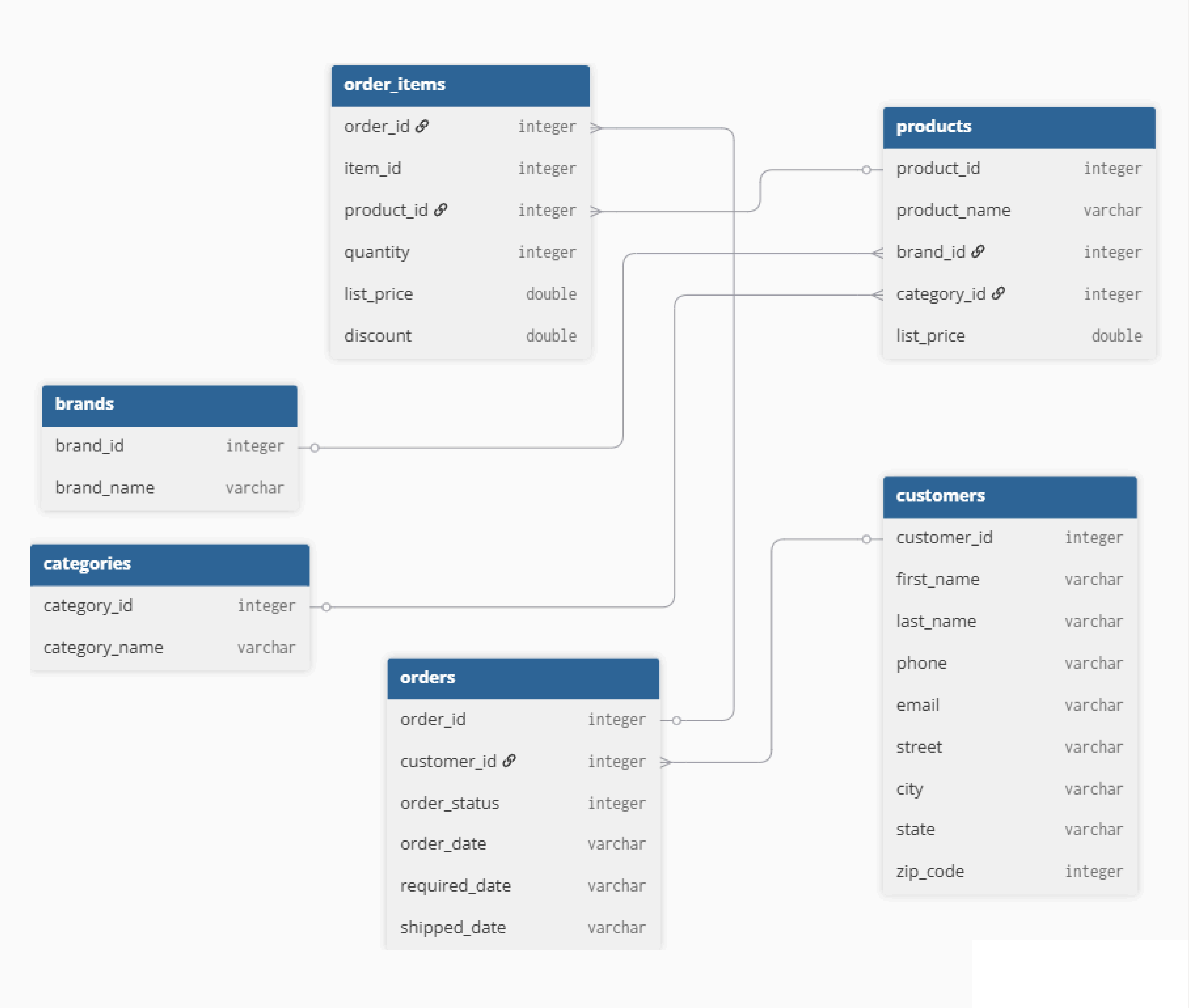
Fiscal Year

FY2016

FY2017

FY2018

Relationship Diagram



Years with most orders

Which year had the highest number of orders?

```
select year(order_date)  Order_Year ,
count(*) Total_Orders
from orders group by Order_Year;
```

Order_Year	Total_Orders
2016	635
2017	688
2018	292

- 2017 received the highest number of customer orders.
- 2018 shows fewer orders due to incomplete available data.
- Business performed better in 2017 than previous years.

Show the most expensive product in every category

```sql
With category as (
select c.Category_name Category ,
p.Product_name Product , p.list_price Price ,
row_number() over(PARTITION BY c.category_name
ORDER BY p.list_price DESC) AS Row_no
from products p join categories c
on p.category_id=c.category_id)
select Category , Product , Price
from category where Row_no=1;
```

Category	Product	Price
Children Bicycles	Electra Townie 3i EQ (20-inch) - Boys' - 2017	489.99
Comfort Bicycles	Electra Townie Go! 8i - 2017/2018	2599.99
Cruisers Bicycles	Electra Townie Commute Go! - 2018	2999.99
Cyclocross Bicycles	Trek Boone 7 Disc - 2018	3999.99
Electric Bikes	Trek Powerfly 8 FS Plus - 2017	4999.99
Mountain Bikes	Trek Remedy 9.8 - 2017	5299.99
Road Bikes	Trek Domane SLR 9 Disc - 2018	11999.99

- Every category has one premium-priced flagship product.
- Premium products help understand each category's price range.
- Management can target premium customers using these products.

Top 10 Customers by Revenue

Who are our most valuable customers ?

```
With customers as
(select c.customer_id,concat(first_name," ",last_name) Customer_Name ,
i.list_price , i.quantity , i.discount from customers c
join orders o on c.customer_id=o.customer_id
join order_items i on o.order_id=i.order_id)
select Customer_Name , round(sum(((1-discount)*list_price)*quantity),2) Total
from customers group by customer_name order by total desc limit 10;
```

Customer_Name	Total
Sharyn Hopkins	34807.94
Pamelia Newman	33634.26
Abby Gamble	32803.01
Lyndsey Bean	32675.07
Emmitt Sanchez	31925.89
Melanie Hayes	31913.69
Debra Burks	27888.18
Elinore Aguilar	25636.45
Corrina Sawyer	25612.7
Shena Carter	24890.62

- A few customers contribute a large share of revenue.
- These customers should receive loyalty rewards and special offers.
- Retaining high-value customers can increase long-term revenue.

Finance department wants customer segmentation.

Classify customers as VIP, Regular, or New.

```
select concat(c.first_name , " " , c.last_name) Customer_name ,
round(sum(((1-i.discount)*i.list_price)*i.quantity),2) Total ,
CASE
        WHEN sum(((1-i.discount)*i.list_price)*i.quantity) <=  10000 THEN 'New'
        WHEN sum(((1-i.discount)*i.list_price)*i.quantity) >= 30000 THEN 'VIP'
        ELSE 'Regular'
    END AS Category
from customers c
join orders o on c.customer_id=o.customer_id
join order_items i on o.order_id=i.order_id
group by Customer_name
order by Total desc;
```

Customer_name	Total	Category
Sharyn Hopkins	34807.94	VIP
Pamelia Newman	33634.26	VIP
Abby Gamble	32803.01	VIP
Lyndsey Bean	32675.07	VIP
Emmitt Sanchez	31925.89	VIP
Melanie Hayes	31913.69	VIP
Debra Burks	27888.18	Regular
Elinore Aguilar	25636.45	Regular
Corrina Sawyer	25612.7	Regular
Shena Carter	24890.62	Regular
Abram Copeland	24607.03	Regular
Neil Mccall	9919.33	New
Catarina Mendez	9908.46	New
Rico Salas	9874.65	New
Vashti Rosario	9865.47	New
Erna Sloan	9849.07	New
Iola Rasmussen	9838.45	New
Ken Charles	9833.65	New
Titus Bullock	9805.25	New

- Customers are grouped based on their spending behavior.
- VIP customers deserve exclusive offers and premium services.
- New customers need marketing campaigns to encourage repeat purchases.

Product Revenue Ranking

Which products generate the most revenue?

```
select p.product_name , round(sum(((1-i.discount)*i.list_price)*i.quantity),2) Revenue
from products p join order_items i on p.product_id=i.product_id
group by p.product_name order by Revenue desc;
```

product_name	Revenue
Trek Slash 8 27.5 - 2016	555558.61
Trek Conduit+ - 2016	389248.7
Trek Fuel EX 8 29 - 2016	368472.73
Surly Straggler 650b - 2016	226765.55
Trek Domane SLR 6 Disc - 2017	211584.62
Surly Straggler - 2016	203507.62
Trek Remedy 29 Carbon Frameset - 2016	203380.87
Trek Powerfly 8 FS Plus - 2017	188249.62
Trek Madone 9.2 - 2017	175899.65
Trek Silque SLR 8 Women's - 2017	174524.73
Trek Silque SLR 7 Women's - 2017	163079.73
Trek Fuel EX 9.8 27.5 Plus - 2017	159052.7
Electra Townie Original 7D EQ - 2016	155169.41

- The highest-selling product generated over $600,000 in revenue.
- Top-ranked products contribute the majority of product sales.
- Low-ranked products may require pricing or marketing improvements.

Most Expensive Product in Every Category

Which premium product represents each category?

```sql
With category as (select c.Category_name Category ,
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- Every category has one premium-priced flagship product.
- Premium products help understand each category's pricing strategy.
- Management can target luxury buyers using these products.

Sales Summary by Brand

Give management a summary of every brand.

```
select brand_name Brand ,
COUNT(p.product_id) AS Total_Products,
round(avg(list_price),2) Avgerage_Price ,
max(list_price) Highest_Price ,
min(list_price) Lowest_Price
from brands b join products p
on b.brand_id=p.brand_id group by brand order by Brand;
```

Brand	Total_Products	Avgerage_Price	Highest_Price	Lowest_Price
Electra	118	761.01	2999.99	269.99
Haro	10	621.99	1469.99	209.99
Heller	3	2173	2599	1320.99
Pure Cycles	3	442.33	449	429
Ritchey	1	749.99	749.99	749.99
Strider	3	209.99	289.99	89.99
Sun Bicycles	23	524.47	1559.99	109.99
Surly	25	1331.75	2499.99	469.99
Trek	135	2500.06	11999.99	149.99

- Brand prices range from around $750 to over $5,000 on average.
- Some brands offer many products, while others focus on fewer premium products.
- Brand comparison helps optimize pricing and inventory decisions.

Compare Year-over-Year growth in revenue.

```
With Year as (select
year(order_date) Year ,
round(sum(((1-discount)*list_price)*quantity),2) Revenue
from orders o join order_items i
on o.order_id=i.order_id group by year)
select Year , Revenue ,
concat(round(((Revenue - Lag(Revenue) Over(Order By Year)) /
Lag(Revenue) Over(Order by Year)) * 100,2)," %") YoY_Growth
from year;
```

Year	Revenue	YoY_Growth
2016	2427378.53	NULL
2017	3447208.24	42.01 %
2018	1814529.79	-47.36 %

- Revenue increased by about 7.8% in 2017 compared to 2016.
- Revenue appears to decline in 2018, but the dataset is incomplete.
- Fair comparison should use only the months available across all years.

Which quarter performs the best?

Which quarter contributes the highest revenue so we can allocate budgets more effectively?

```
select
concat('Q',quarter(order_date)) Quarter ,
round(sum(((1-discount)*list_price)*quantity),2) Revenue
from orders o join order_items i
on o.order_id=i.order_id group by quarter
order by revenue desc;
```

Quarter	Revenue
Q1	2405390.36
Q2	2275476.84
Q3	1540939.58
Q4	1467309.77

- Quarter 2 generated the highest revenue, making it the peak sales period.
- Quarter 1 recorded the second-highest business performance.
- Inventory and marketing should be strengthened before Quarter 2.

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on b.brand_id=p.brand_id group by brand order by Brand;
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- Premium brands maintain higher average selling prices than others.
- Some brands focus on affordable products with larger product portfolios.
- Brand-level insights support pricing and sales strategy decisions.

Customers with More Than One Order

Identify repeat customers.

```
select  concat(first_name," ",last_name) Cust_Name ,
count(*) Total
from orders o join customers c
on o.customer_id=c.customer_id
group by o.customer_id , c.first_name , c.last_name
having total > 1 order by total desc limit 10;
```

Cust_Name	Total
Cleotilde Booth	3
Genoveva Baldwin	3
Latasha Hays	3
Robby Sykes	3
Monika Berg	3
Mozelle Carter	3
Araceli Golden	3
Williemae Holloway	3
Corene Wall	3
Lashawn Ortiz	3

- Several customers placed 2 to 5 orders, showing repeat purchases.
- Repeat customers indicate customer satisfaction and brand loyalty.
- Retaining repeat customers is more cost-effective than acquiring new ones.

Month-wise Orders Comparison

```
select year(order_date) Year ,
monthname(order_date) as Month ,
count(*) Total_orders
from orders where month(order_date)<5
group by year(order_date) ,
month(order_date),monthname(order_date)
order by year(order_date);
```

Year	Month	Total_orders
2016	January	50
2016	February	49
2016	March	55
2016	April	43
2017	January	50
2017	February	57
2017	March	67
2017	April	57
2018	January	52
2018	February	35
2018	March	68
2018	April	125

- Comparing January to April provides a fair comparison across all years.
- Order volume in April'18 shows sudden spike.
- Missing data after April explains the apparent yearly decline.

Thanks for your attentive analysis

Khushal Dusane

Data Analyst

SQL | Python | Power BI | Excel

[Download Project here]

khushaldusane2003@gmail.com | 9730116026